

Israt Jahan Mou

AI / ML Engineer · Machine Learning Researcher

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PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering undergraduate (CGPA 3.78 / 4.00) with hands-on experience in machine learning, deep learning, computer vision, explainable AI, and applied research. Built and deployed end-to-end ML systems for employee attrition prediction and IoT intrusion detection – covering model evaluation, imbalanced learning, and explainability – and engineered PyTorch-based deep learning pipelines (YOLOv11, Video Swin Transformer) for IoT security and real-time surveillance threat detection. Every model I ship is paired with explainability checks, robustness tests, and a deployment target.

EDUCATION

B.Sc. in Computer Science and Engineering

Expected Graduation: 2027

University of Asia Pacific (UAP), Dhaka, Bangladesh

CGPA: **3.78** / 4.00 · Specialization: ML, DL & Computer Vision · Research output: IEEE manuscript in preparation.

TECHNICAL SKILLS

Programming: Python, Java, SQL, JavaScript, HTML/CSS

Machine Learning: Scikit-learn, Linear & Logistic Regression, KNN, Naive Bayes, Decision Trees, Random Forest, SVM, Ensemble Learning, Classification, Regression, Clustering, Feature Engineering, Feature Scaling, Dimensionality Reduction, Imbalanced Learning, SMOTE, Cross-Validation, Model Evaluation, Hyperparameter & Threshold Tuning

Deep Learning & Computer Vision: PyTorch, Neural Networks, Transfer Learning, CNNs, Transformer-based Models, YOLOv11, Video Swin Transformer, Object Detection, Spatio-Temporal Modeling

Explainable AI & Security: SHAP, Explainable ML, Adversarial Robustness (FGSM), Intrusion Detection

Data, Tools & Deployment: NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, Jupyter, Google Colab, Kaggle, VS Code, Joblib, Streamlit, Gradio, Hugging Face Spaces, Django, Django REST Framework, React, SQLite

RESEARCH & SELECTED AI/ML PROJECTS

MOI-Lite + E-SATF: Lightweight Hierarchical IDS for IoT

2026

Deep Learning, SHAP, TON-IoT, UNSW-NB15, CIC-IDS2017

- Designed MOI-Lite, a lightweight two-stage IDS combining multi-scale dilated 1D-CNN blocks with lightweight self-attention for malicious IoT traffic classification; evaluated cross-dataset generalization and model compression across TON-IoT, UNSW-NB15, and CIC-IDS2017.
- Achieved **99.79% binary F1** and **94.37% multiclass macro F1**; INT8 quantization reduced the model footprint to **65 KB** for resource-constrained deployment.
- Developed E-SATF explanation-stability regularization for SHAP consistency; evaluated FGSM adversarial robustness, McNemar statistical comparisons, and inference latency. *Manuscript in preparation for IEEE submission.*

Multi-Stage Hybrid Architecture for Real-Time Surveillance Threat Detection

2026

B.Sc. Final-Year Thesis – YOLOv11 + Video Swin Transformer + Risk Fusion

- Engineered a multi-stage deep-learning pipeline combining YOLOv11 (weapon / object detection) with Video Swin Transformer (spatio-temporal abnormal-behavior recognition) for real-time surveillance threat detection. Designed a risk-fusion mechanism that combines spatial and behavioral signals into graded threat assessments, supporting real-time triage instead of binary alerts.

Employee Attrition Risk Predictor

2026

Python, Scikit-learn, SMOTE, SHAP, Streamlit

- Built and deployed an end-to-end ML system on the IBM HR Analytics dataset (1,470 records) covering preprocessing, imbalanced learning, model comparison, explainability, and deployment.
- Built a reproducible Scikit-learn pipeline (ColumnTransformer, OneHotEncoder, StandardScaler, SMOTE on training data only); compared Logistic Regression, Random Forest, and Decision Tree via stratified 5-fold CV, selecting Logistic Regression for minority-class recall. Optimized the classification threshold via precision–recall analysis and deployed a 30-feature Streamlit app with risk tiers, SHAP LinearExplainer-based explanations, and retention recommendations.

ADDITIONAL SOFTWARE PROJECTS

Glam Girl – Full-Stack eCommerce Platform

2025

Django, Django REST Framework, React, SQLite

- Developed a full-stack eCommerce platform with a Django and Django REST Framework backend and React frontend, including product, cart, and order management.

AI Rasengan: Chakra Control System

2026

JavaScript, MediaPipe Hands, HTML5 Canvas

- Built a browser-based computer-vision interaction system using hand landmarks and gesture classification, with a real-time HTML5 Canvas particle engine.

Text Analyzer Pro & SQL Practice

2026

Python, Tkinter, CLI, Pytest, MySQL

- Desktop / CLI text-analysis application with word-frequency statistics, search, JSON / TXT export, and automated testing; plus a MySQL practice module covering relational schema design, joins, aggregation, and string functions.